

Designing climate-resilient agriculture through crop portfolios

Abstract (1000 characters): Hotter, drier summers and increasing extremes are challenging growers across the world. This is especially clear in Europe where summer heat is increasing faster than in the Arctic. Maintaining agricultural regions that continue to provide economic livelihoods, high-quality products while continuing traditional practices requires understanding of the climate resiliency of current crops, and how to best exploit it. Here I propose to leverage the deep agriculture knowledge, data and cutting-edge research in Montpellier with my skills in hierarchical modeling to address this. Working with unparalleled data across varieties and clones of one major crop, we will quantify heat tolerance across cultivars and clones to help growers develop climate-smart portfolios.

Background

Resilient agricultural systems today must deal with the increasing stressors of climate change. Hotter, drier summers and increasing extremes in heat, cold and precipitation are challenging growers across the world. This is especially true in Europe where summer heat is increasing faster than in the Arctic, and where reports of crop damage and failures are increasing alongside the heat. High temperatures can slow plant development and cause tissue damage couple2024clusters, leading to yield declines and making it harder for growers to maintain croplands. How to develop agricultural regions that continue to provide economic livelihoods and high-quality products while continuing traditional practices with climate change is one of the greatest challenges of the next century. Meeting this challenge requires a greater understanding of the climate resiliency of current crops, and how to best exploit it as the climate continues to shift.

One of the most effective methods for climate resiliency is to exploit the diversity of existing climate tolerance in current crops. Different crop species and different cultivars within species require and can tolerate different levels of heat. Matching this diversity to future climates can thus allow growers to maintain many of their traditional practices and plant the same species well into the future—depending on the climate diversity of that crop species, and how well we can estimate it.

Progress in forecasting when growers should change their planting regimes with climate change has been slow because of our limited knowledge of the temperature response of different crops. Fundamental temperature response curves (FIG) underlie the biology of all organisms and are critical to forecasting crops with climate change. These models usually capture the non-linear effect of temperature, where increasing temperatures beyond some fundamental minimum (below which cells cannot grow) generally accelerate development and growth until some maximum rate—which defines the optimal temperature. At temperatures higher than this optimum, development and growth slows and then stalls, driving crop failures.

Estimating temperature responses curves—and their critical values of the temperature minimum, optimum and maximum—is extremely difficult. To date most estimates of complete temperature response curves exist for only a handful of herbaceous crops that are amenable to small-scale experiments (generally done in controlled environment chambers where temperatures are held constant and rates can be carefully measured). Heat tolerance estimates in many species, and effectively all woody species, rely on modeling approaches that cannot robustly estimate temperature maxima or minima (and thus they are usually fixed) and where estimates are easily biased by data quality issues (e.g., far more data for one species than another).

Today, many models of future food security assume that growers will need to shift their crops over time with increasing climate change, but generally only can forecast major shifts—for example, from winegrapes to avocados—that would require immense human adaptation and the likely loss of

traditional practices. Improved estimates of the temperature response of different crop species and their cultivars could rapidly help growers develop more climate resilient systems. Such information would allow growers to exploit the full climate diversity of their existing crop species, before considering a major shift.

Estimates across a large number of cultivars would also help growers develop climate resilient portfolios of cultivars—selecting the best mix of cultivars for each grower given their preferences and region. A portfolio of cultivars can better handle climate variation year to year, as several may perform best in slightly cooler years and several others in slightly warmer years. At the same time, adjusting the mixture over time can help adapt to long-term trends due to anthropogenic climate change. The concept of a careful portfolio of climatically matched cultivars is not novel, but how much it is used depends on local practices and the particular crop. It also depends on growers having the information they need on how different cultivars respond to climate.

What?

Working with colleagues in Montpellier, I will leverage new methods alongside newly available genetic and development data to estimate the climate resiliency of a major crop—winegrapes. We plan to use winegrapes as a case study because they currently have the data on seasonal development and genetic information required to build the most robust models, but our approach will be specifically designed to extend to other crops.

Our approach will overcome the major data quality issues that have stymied previous efforts to estimate full temperature response curves across cultivars—called varieties in winegrapes—through multi-level (also known as hierarchical) modeling. This approach estimates at once an overall species-level temperature response curve at the same time it estimates differences across varieties, while accounting for uneven sampling and spatial variation. Highly uneven data across cultivars is a common problem in crops, including winegrapes, because a few varieties are often very popular, widely planted and have extensive data (e.g., Chardonnay, Pinot noir), while other varieties are less common but may harbor excellent heat tolerance or other temperature response attributes that could be make them ideal for adapting to climate change (e.g., Xinomavro). I have developed new modeling approaches CITENACHO that can handle this uneven sampling by incorporating genetic relatedness information on cultivars to better estimate each variety. They also can be adapted to include additional layers of genetic information. For example, within each winegrapes variety multiple clones exist (where each clone has unique somatic mutation differences) and this can be further incorporated into the models to better understand climate resiliency within each variety and clone, allowing growers to fine-tune their portfolios.

This approach relies on the extensive data and expertise of my colleagues in Montpellier, who have recently helped lead a major new analysis of the genetic relatedness of winegrape varieties, which we leverage here. They also provide a depth of knowledge of traditional practices and current challenges for winegrowing in France, making the proposed work better positioned to directly aid growers. Additionally, they have seasonal development data required to estimate temperature response curves (described further below) observed over high temperatures. The lack of observational data at high temperatures has limited previous work to estimate optimum and maximum temperatures for crops, but recent heat extremes in Montpellier have pushed plants into these regions. My colleagues have been on the forefront of documenting this couple2024clusters and provide rare data and expertise we will leverage. This project thus brings together my modeling expertise, with the expertise on winegrape genetic diversity, response to heat extremes and critical regional and larger-scale understanding of winegrowing.

How?

Our approach centers around (1) the domestication of winegrapes across Europe’s extensive climate gradient that led to cultivars (varieties) that likely vary strongly in their heat tolerance and (2) the shared evolutionary history of winegrape cultivars that we can use for more robust estimates across diverse varieties. We will accomplish this by leveraging existing climate data with Domaine de Vassal’s incredible collection of seasonal development data across over 200 varieties, alongside likely other data from resources across INRAe and other regional programs that have collected data on season development, from budburst to flowering, and fruit ripening (veraison).

Collaborators in Montpellier will provide expertise in the seasonal development data across cultivars and their genetic relationships, which we will model by combining two unique modeling approaches my lab has developed. The first is a fully differentiable thermal development curve (Fig. fig:tempcurve), and the second is a using an hierarchical approach that leverages genetic relationships between cultivars to best estimate the heat tolerance of each cultivar. Each approach has successfully been applied in other cases to woody plants, but have not been combined to estimate thermal responses across diverse cultivars. Given sufficient information on clones (within cultivars) we will additionally add this hierarchical level to separate out the climate adaptation potential of clones versus cultivars.

Timeline: Data gathering is already underway, with the goal of starting work on model design and parameterization in March 2028 when I arrive in France. In my first month I plan meetings to review and refine my modeling plan with my collaborators, who bring unique expertise in variety diversity, its genetics and how to make results most immediately useful to growers. Model development and fitting will be an iterative process focused on design, parameterization, validating and eventually forecasting. I expect continual consultations with my French collaborators and likely two months of discussion and effort for fitting and validation. In May I hope to begin using the model to prepare historical results, followed by forecasting with the model in June and finalizing results in July.

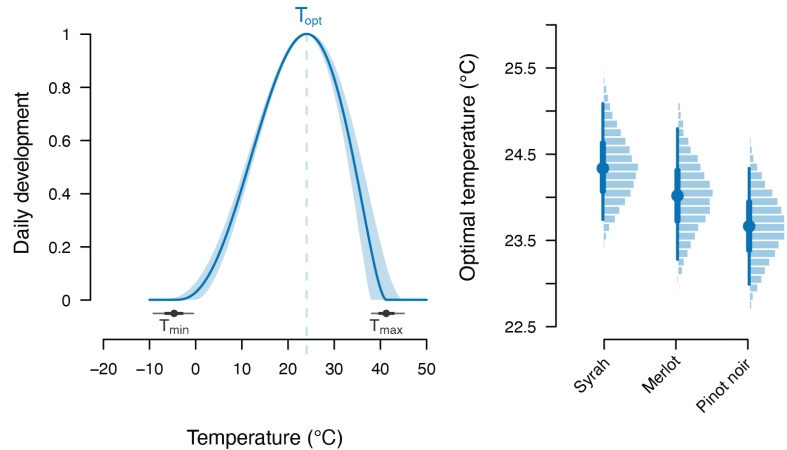


Figure 1: This project will estimate a temperature response curve for winegrapes as a crop, including heat tolerance variation across cultivars. Here we show current results for veraison thermal development (left) and variation in the optimal temperature for three cultivars (right) but this project will extend to up to 200 different cultivars (exact number depends on data for each phenological stage) and more accurately model them given genetic information.

Impact, outcomes & benefits

Our aim is to provide one major building block to climate-smart agriculture: robust estimates of the thermal tolerance of a crop, and its underlying genetic levels, to help growers develop a portfolio approach to future climate. This information should allow growers to build portfolios of varieties

that can deal with current climate means and extremes, and help them decide when to shift this portfolio as the climate shifts in the future.

My research on winegrapes to date has generated high interest, both in research and public spheres. For example, my lab's recent work on variety change with warming [2], has been cited almost 50 times in one year and led to dozens of radio and print stories, including by the BBC, NPR, *The Guardian*, *Washington Post*, *El Mundo* and France24 News. I plan to communicate results through publications (I expect at least one paper in *Nature Climate Change* or similar, one methods and conceptual paper in a high-profile climate change focused journal), press releases and then at various conferences and seminar series I am involved with, including both in academic (e.g., universities), grower-focused (e.g., Napa County Farm Bureau, BC Winegrape Council) and mixed venues (e.g., international Cool Climates Wine Symposium).

Importantly, I expect this research will also open new lines of communication. Climate change is a global challenge and solutions will come from shared problem solving and resilience. My hope is this research will help develop new links with growers and researchers in Europe and in turn foster deeper connections with their counterparts in North America.

Give a talk in Montpellier? Or other outreach opportunities?

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- [1] Coupel-Ledru, A., Westgeest, A. J., Albasha, R., Millan, M., Pallas, B., Doligez, A., Flutre, T., Segura, V., This, P., Torregrosa, L., et al. (2024). Clusters of grapevine genes for a burning world. *The New phytologist*.
 - [2] Morales-Castilla, I., de Cortazar-Atauri, I. G., Cook, B. I., Lacombe, T., Parker, A., van Leeuwen, C., Nicholas, K. A., and Wolkovich, E. M. (2020). Diversity buffers winegrowing regions from climate change losses (vol 117, pg 2864, 2020). *Proceedings of the National Academy of Sciences of the United States of America* *117*, 30860–30860.